CHAPTER ONE

INTRODUCTION

1.1 Definition of economics

Economics is one of the most exciting disciplines in social sciences. The word economy comes from the Greek phrase ―one who manages a household‖. *Adam Smith* – generally known as the father of economics – brought out his famous book, ―*An Inquiry into the Nature and Causes of Wealth of Nations*‖, in the year 1776.

Economics is a social science which studies about efficient allocation of scarce resources so as to attain the maximum fulfillment of unlimited human needs.

- ❖ Economics is a science of choice
- ❖ It studies how people choose to use scarce or limited productive resources (land, labor, equipment, technical knowledge and the like) to produce various commodities.

1.2 The rationales of economics

There are two fundamental facts that provide the foundation for the field of economics.

1) Human (society‘s) material wants are unlimited.

2) Economic resources are limited (scarce).

The **basic economic problem** is about **scarcity** and **choice** since there are only limited amount of resources available to produce the unlimited amount of goods and services we desire.

1.3 Scope and method of analysis in economics

1.3.1 Scope of economics

The scope of modern economics is formed by its two major branches:

- ❖ Microeconomics and
- ❖ Macroeconomics

That means economics can be analyzed at micro and macro level.

A. Microeconomics is concerned with the economic behavior of individual decision making units such as households, firms, markets and industries. In other words, it deals with how households and firms make decisions and how they interact in specific markets.

- ✓ Studies individual economic units of an economy.
- ✓ Deals with individual income, individual prices, individual outputs, etc.
- ✓ Its central problem is price determination and allocation of resources.
- ✓ Its main tools are the demand and supply of particular commodities and factors.

Examples: Individual income, individual savings, individual prices, an individual firm‘s output, individual consumption, etc.

B. Macroeconomics is a branch of economics that deals with the effects and consequences of the aggregate behavior of all decision making units in a certain economy. It looks economy as a whole and discusses about the economy-wide phenomena

- ✓ Studies an economy as a whole and its aggregates.
- ✓ Deals with national income and output and general price level
- ✓ Its central problem is determination of level of income and employment.
- ✓ Its main tools are aggregate demand and aggregate supply of an economy

Examples: national income, national savings, general price level, national output, aggregate consumption, etc.

1.3.2 Positive and normative analysis

Economics can be analyzed from two perspectives:

- ✓ Positive economics and
- ✓ Normative economics.

Positive economics: it is concerned with analysis of **facts** and attempts to describe the world as it is. It tries to answer the questions what was; what is; or what will be? It **does not judge a** system as good or bad, better or worse.

Example:

- The current inflation rate in Ethiopia is 12 percent.
- Poverty and unemployment are the biggest problems in Ethiopia.
- The life expectancy at birth in Ethiopia is rising.

Any disagreement on positive statements can be checked by looking in to facts.

Normative economics: It deals with the questions like, what ought to be? Or what the economy should be? It evaluates the desirability of alternative outcomes based on one‘s **value judgments** about what is good or what is bad.

Example:

- The poor should pay no taxes.
- There is a need for intervention of government in the economy.
- Females ought to be given job opportunities.

Any disagreement on a normative statement can be solved by voting

1.3.3 Inductive and deductive reasoning in economics

There are two methods of logical reasoning: inductive and deductive.

a) Inductive reasoning is a logical method of reaching at a correct general statement or theory based on several independent and specific correct statements. In short, it is the process of deriving a principle or theory by moving from **facts to theories** and from **particular to general** economic analysis.

b) Deductive reasoning is a logical way of arriving at a particular or specific correct statement starting from a correct general statement. The theory may agree or disagree with the real world and we should check the validity of the theory to facts by moving from **general to particular**

1.4 Scarcity, choice, opportunity cost and production possibilities frontier

1. Scarcity

The fundamental economic problem that any human society faces is the problem of **scarcity**.

Scarcity refers to the fact that all economic resources that a society needs to produce goods and services are finite or limited in supply.

Thus, the term scarcity reflects the imbalance between our wants and the means to satisfy those wants.

2Resource

Free resources: A resource is said to be free if the amount available to a society is greater than the amount people desire at zero price.

E.g. sunshine

Scarce (economic) resources: A resource is said to be scarce or economic resource when the amount available to a society is less than what people want to have at zero price.

Economic resources are usually classified into four categories.

- **Labour:** refers to the physical as well as mental efforts of human beings in the production and distribution of goods and services. The reward for labour is called **wage.**
- **Land:** refers to the natural resources or all the free gifts of nature usable in the production of goods and services. The reward for the services of land is known as **rent.**

- **Capital**: refers to all the manufactured inputs that can be used to produce other goods and services. Example: equipment, machinery, transport and communication facilities, etc. The reward for the services of capital is called **interest.**
- **Entrepreneurship**: refers to a special type of human talent that helps to organize and manage other factors of production to produce goods and services and takes risk of making loses. The reward for entrepreneurship is called **profit.**

Entrepreneurs are individuals who:

- Organize factors of production to produce goods and services.
- Make basic business policy decisions.
- Introduce new inventions and technologies into business practice.
- Look for new business opportunities.
- Take risks of making losses.

2. **Choice:** Due to the problem of scarcity, individuals, firms and government are forced to choose as to what output to produce, in what quantity, and what output not to produce. In short, **scarcity implies choice.** Choice, in turn, implies cost. That means whenever choice is made an alternative opportunity is sacrificed. This cost is known as opportunity cost.

Scarcity → limited resource → limited output → we might not satisfy all our wants

→choice involves costs → opportunity cost

3. **Opportunity cost:** Opportunity cost is the amount or value of the next best alternative that must be sacrificed (forgone) in order to obtain one more unit of a product.

For example, suppose you have Br.10 and are deciding how to spend it. Should you buy five exercises books? A ticket for cinema? Two large fast foods? Note, however, that once you decided to do one of these alternatives with your money (Br.10), the other alternatives are sacrificed. Therefore, opportunity cost of any decision is a sacrifice made in choosing the alternative.

4. The Production Possibilities Frontier or Curve (PPF/ PPC)

The production possibilities frontier (PPF) is a curve that shows the various possible combinations of goods and services that the society can produce given its scarce resources and technology. To draw the PPF we need the following assumptions.

a. The quantity as well as quality of economic resource available for use during the year is fixed.
b. There are two broad classes of output to be produced over the year.
c. The economy is operating at full employment and is achieving full production (efficiency).

d. Technology does not change during the year.

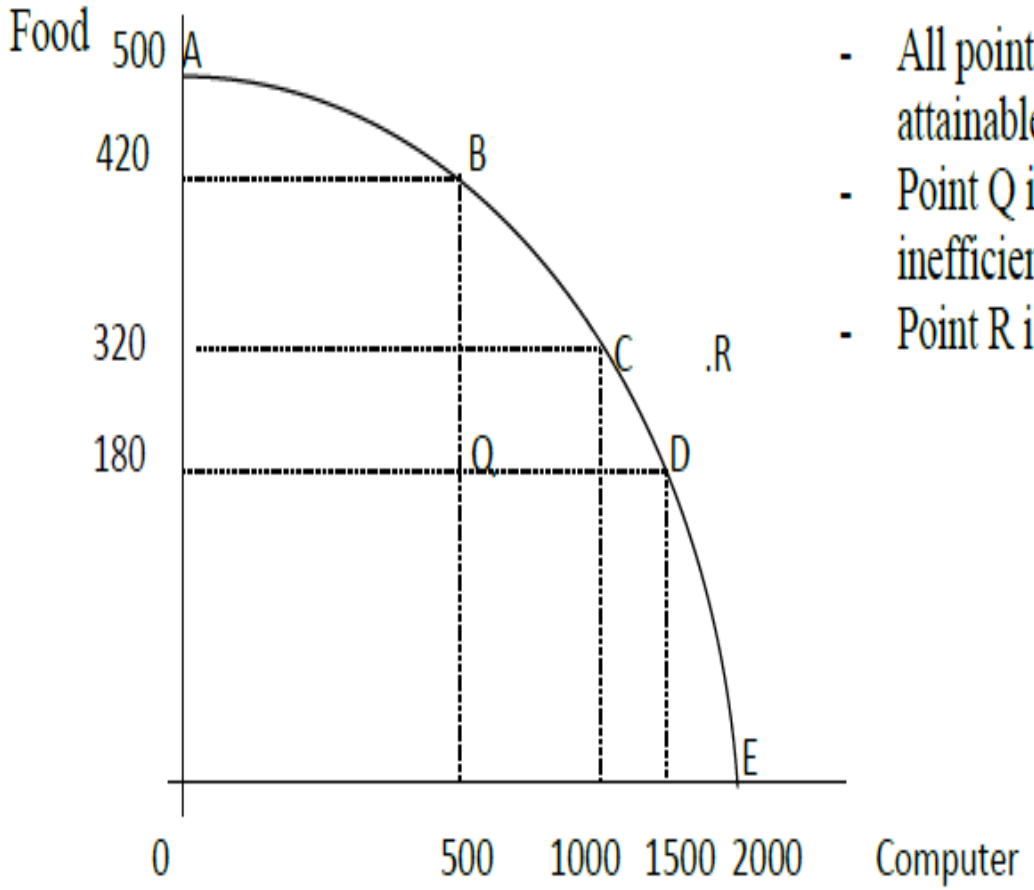
e. Some inputs are better adapted to the production of one good than to the production of the other (specialization).

Suppose a hypothetical economy produces food and computer given its limited resources and available technology (table 1.1).

Table 1.1: Alternative production possibilities of a certain nation

Types of product	Unit	Production alternatives				
		A	B	C	D	E
Food	Metric ton	500	420	320	180	0
Computer	Number	0	500	100	1500	200

We can also display the above information with a graph.



- All points on the PPF are attainable and efficient
- Point Q is attainable but inefficient
- Point R is unattainable

Figure 1.1: Production Possibilities Frontier

Summary:

- Points on the curve are both attainable and efficient
- Points inside the curve are attainable but inefficient
- Points outside the curve are unattainable

To sum up, PPF illustrates four basic concepts:

- **Scarcity of resources: -** this is reflected by the negative slope of the PPF. Moreover, points outside the curve are unattainable because of the scarcity of resources.
- **Choice: -** this is reflected in the need for the society to select among the various attainable goods on the curve. **Downward slope of the PPF: -** this indicates the trade-offs that exist in the real world, i.e. opportunity cost.
- **Law of increasing opportunity cost: -** this is reflected by the concavity of the PPF

$$Opportunity\ cost\ of\ a\ good = \frac{\text{the amount of the next best alternative sacrified}}{\text{the amount of the good gained}}$$

Example: Referring to table 1.1 above, if the economy is initially operating at point B, what is

the opportunity cost of producing one more unit of computer?

Solution: Moving from production alternative B to C we have

$$OC = \frac{320-420}{1000-500} = \left|\frac{-100}{500}\right| = \underline{\underline{0.2}}$$ (The economy gives up 0.2 metric tons of food per computer)

Economic Growth and the PPF

Economic growth or an increase in the total output level occurs when one or both of the following conditions occur.

1. Increase in the quantity or/and quality of economic resources.

2. Advances in technology.

Economic growth is represented by outward shift of the PPF.

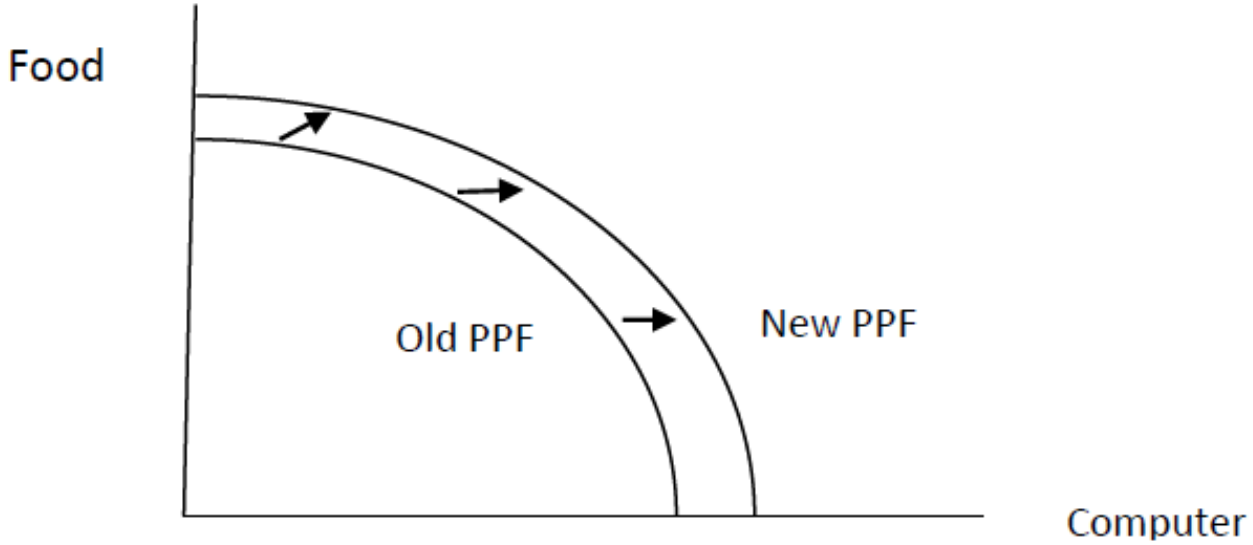


Figure 1.2: Economic growths with a new PPC

An economy can grow because of an increase in productivity in one sector of the economy. For example, an improvement in technology applied to either food or computer would be illustrated by a shift of the PPF along the Y- axis or X-axis. This is called *asymmetric growth* (figure 1.3).

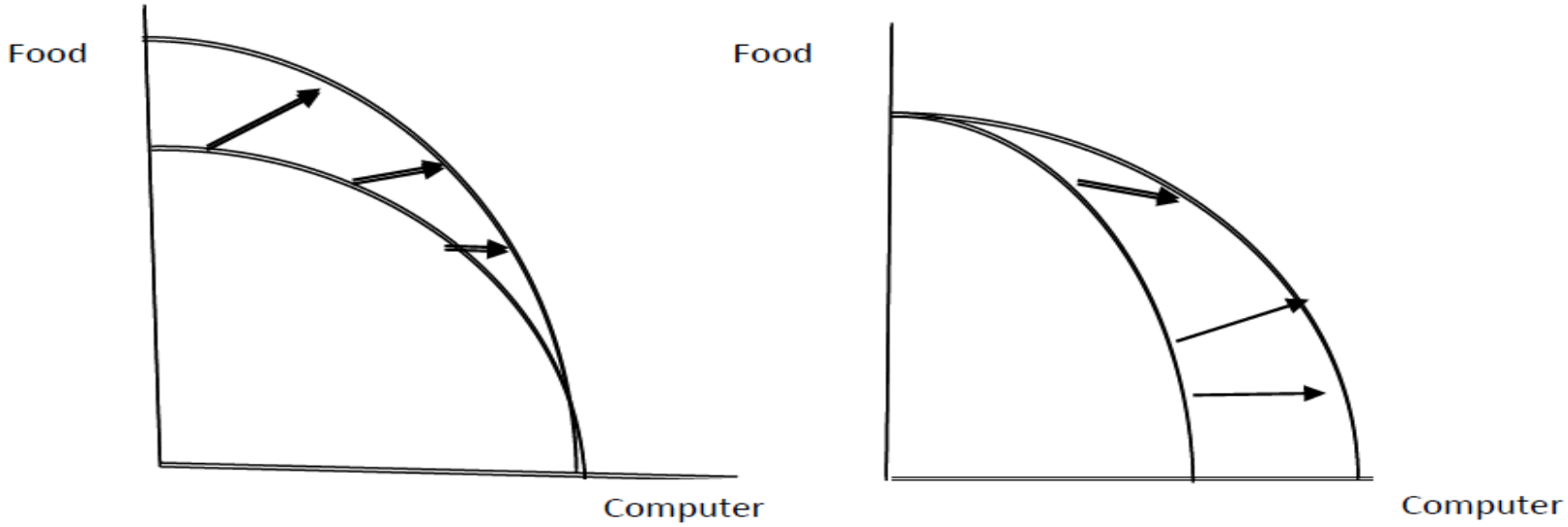


Figure 1.3: improvements in technology and quantity and/or quality of resources on national output

1.5 Basic economic questions

Economic problems faced by an economic system due to scarcity of resources are known as basic economic problems. Therefore, any human society should answer the following three basic questions.

What to Produce?

This problem is also known as the *problem of allocation of resources*. The economy must make choices such as

- ❖ Consumption goods versus capital goods,
- ❖ Civil goods versus military goods, and
- ❖ Necessity goods versus luxury goods.

How to Produce?

This problem is also known as the *problem of choice of technique*.

Broadly speaking, the various techniques of production can be classified into two groups: *labour-intensive techniques* and *capital-intensive techniques*.

- A labour-intensive technique involves the use of more labour relative to capital, per unit of output.
- A capital-intensive technique involves the use of more capital relative to labour, per unit of output.

The choice between different techniques depends on the available **supplies of different factors of production and their relative prices**.

For Whom to Produce?

This problem is also known as the *problem of distribution of national product*. It relates to how a material product is to be distributed among the members of a society. The economy must decide, for example, whether to produce for the benefit of the few rich people or for the large number of poor people.

1.6 Economic systems

The way a society tries to answer the above fundamental questions is summarized by a concept known as economic system.

- An **economic system** is a set of organizational and institutional arrangements established to answer the basic economic questions. Customarily, we can identify three types of economic system. These are:
- Capitalism,
- Command and
- Mixed economy.

1.6.1 Capitalist economy

- *Capitalism* is the oldest formal economic system in the world.
- It became widespread in the middle of the 19th century.
- In this economic system, all means of production are **privately owned**, and production takes place at the initiative of individual private entrepreneurs who work mainly for private profit.
- Government intervention in the economy is minimal.
- This system is also called *free market economy* or *market system* or *laissez faire*.

Features of Capitalistic Economy

The right to private property: Economic or productive factors such as land, factories, machinery, mines etc. are under private ownership.

Freedom of choice by consumers: Consumers can buy the goods and services that suit their tastes and preferences.

Profit motive: Entrepreneurs, in their productive activity, are guided by the motive of profit-making.

Competition: In a capitalist economy, competition exists among sellers or producers of similar goods to attract customers. Among buyers, there is competition to obtain goods.

Among workers, the competition is to get jobs. Among employers, it is to get workers and investment funds.

Price mechanism: All basic economic problems are solved through the price mechanism.

Minor role of government: The government does not interfere in day-to-day economic activities and confines itself to defense and maintenance of law and order.

Self-interest: Each individual is guided by self-interest and motivated by the desire for economic gain.

Inequalities of income: There is a wide economic gap between the rich and the poor.

Existence of negative externalities: A negative externality is the harm, cost, or inconvenience suffered by a third party because of actions by others

Advantages of Capitalistic Economy

- **Flexibility or adaptability:** It successfully adapts itself to changing environments.
- **Decentralization of economic power:** Market mechanisms work as a decentralizing force against the concentration of economic power.
- **Increase in per-capita income and standard of living:** Rapid growth in levels of production and income leads to higher per-capita income and standards of living.
- **New types of consumer goods:** Varieties of new consumer goods are developed and produced at large scale.
- **Growth of entrepreneurship:** Profit motive creates and supports new entrepreneurial skills and approaches.
- **Optimum utilization of productive resources:** Full utilization of productive resources is possible due to innovations and technological progress.
- **High rate of capital formation:** The right to private property helps in capital formation.

Disadvantages of Capitalistic Economy

- **Inequality of income:** Capitalism promotes economic inequalities and creates social imbalance.
- **Unbalanced economic activity:** As there is no check on the economic system, the economy can develop in an unbalanced way in terms of different geographic regions and different sections of society.
- **Exploitation of labour:** In a capitalistic economy, exploitation of labour (for example by paying low wages) is common.
- **Negative externalities:** are problems in capitalistic economy where profit maximization is the main objective of firms. If economic makes sense for a firm to force others to pay the impacts of negative externalities such as pollution.

2. Command economy

Command economy is also known as socialistic economy. Under this economic system, the economic institutions that are engaged in production and distribution are owned and controlled by the state. In the recent past, socialism has lost its popularity and most of the socialist countries are trying free market economies.

Main Features of Command Economy

- ✓ Collective ownership
- ✓ Central economic planning
- ✓ Strong government role
- ✓ Maximum social welfare
- ✓ Relative equality of incomes

3. Mixed economy

A mixed economy is an attempt to combine the advantages of both the capitalistic economy and the command economy. It incorporates some of the features of both allows private and public sectors to co-exist.

- **Main Features of Mixed Economy**
- **Co-existence of public and private sectors**
- **Economic welfare:** Economic welfare is the most important criterion of the success of a mixed economy.
- **Economic planning**
- **Price mechanism**
- **Economic equality**

6.Decision making units and the circular flow model

There are three decision making units in a closed economy. These are households, firms and the government.

i) **Household:** A household can be one person or more who live under one roof and make joint financial decisions. Households make two decisions.

a) Selling of their resources, and

b) Buying of goods and services.

ii) **Firm:** A firm is a production unit that uses economic resources to produce goods and services. Firms also make two decisions:

a) Buying of economic resources

b) Selling of their products.

iii) **Government**: A government is an organization that has legal and political power to control or influence households, firms and markets. Government also provides some types of goods and services known as **public goods and services** for the society.

The three economic agents interact in two markets:

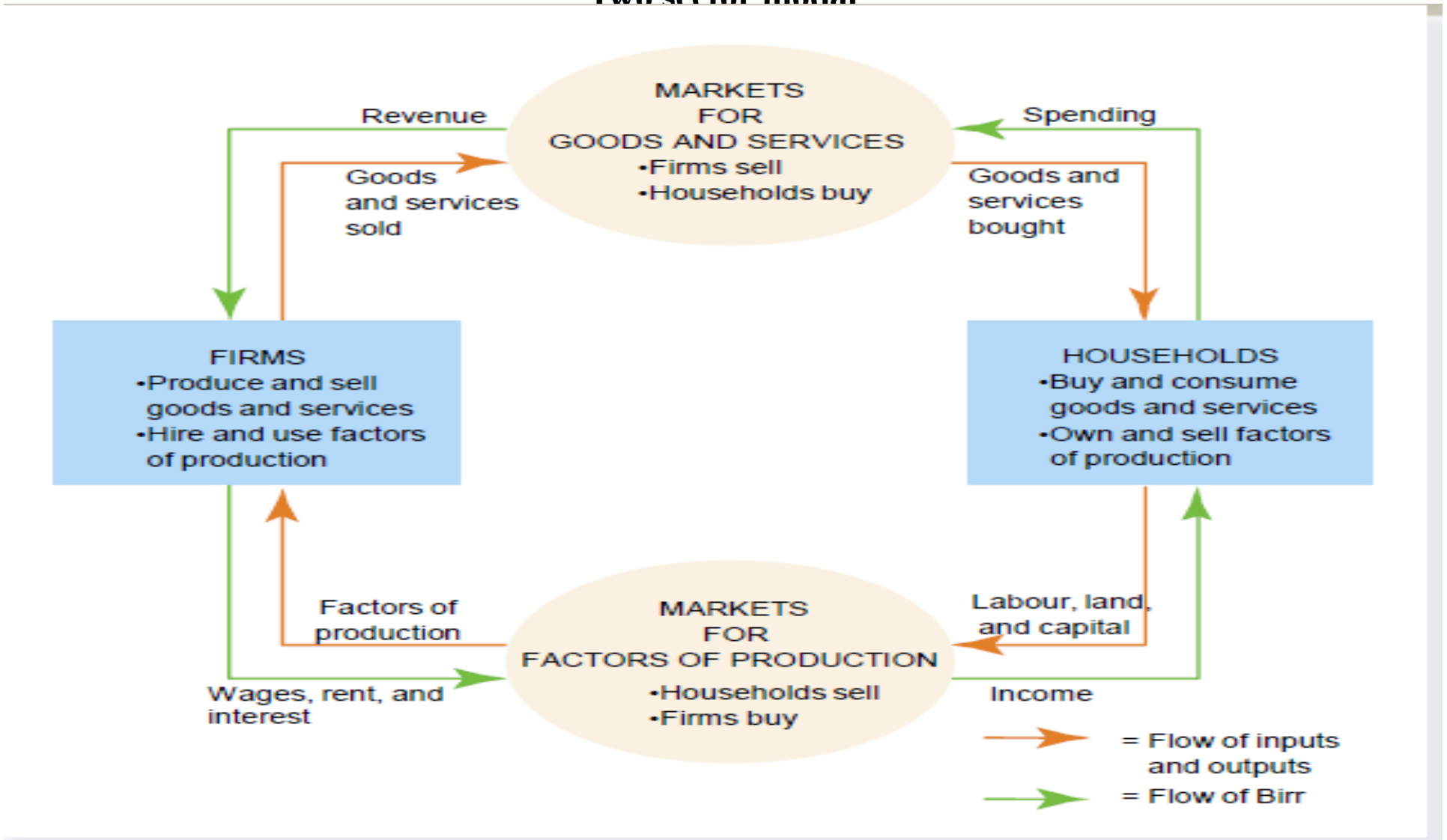
Product market: it is a market where goods and services are transacted/ exchanged.

That is, a market where households and governments buy goods and services from business firms.

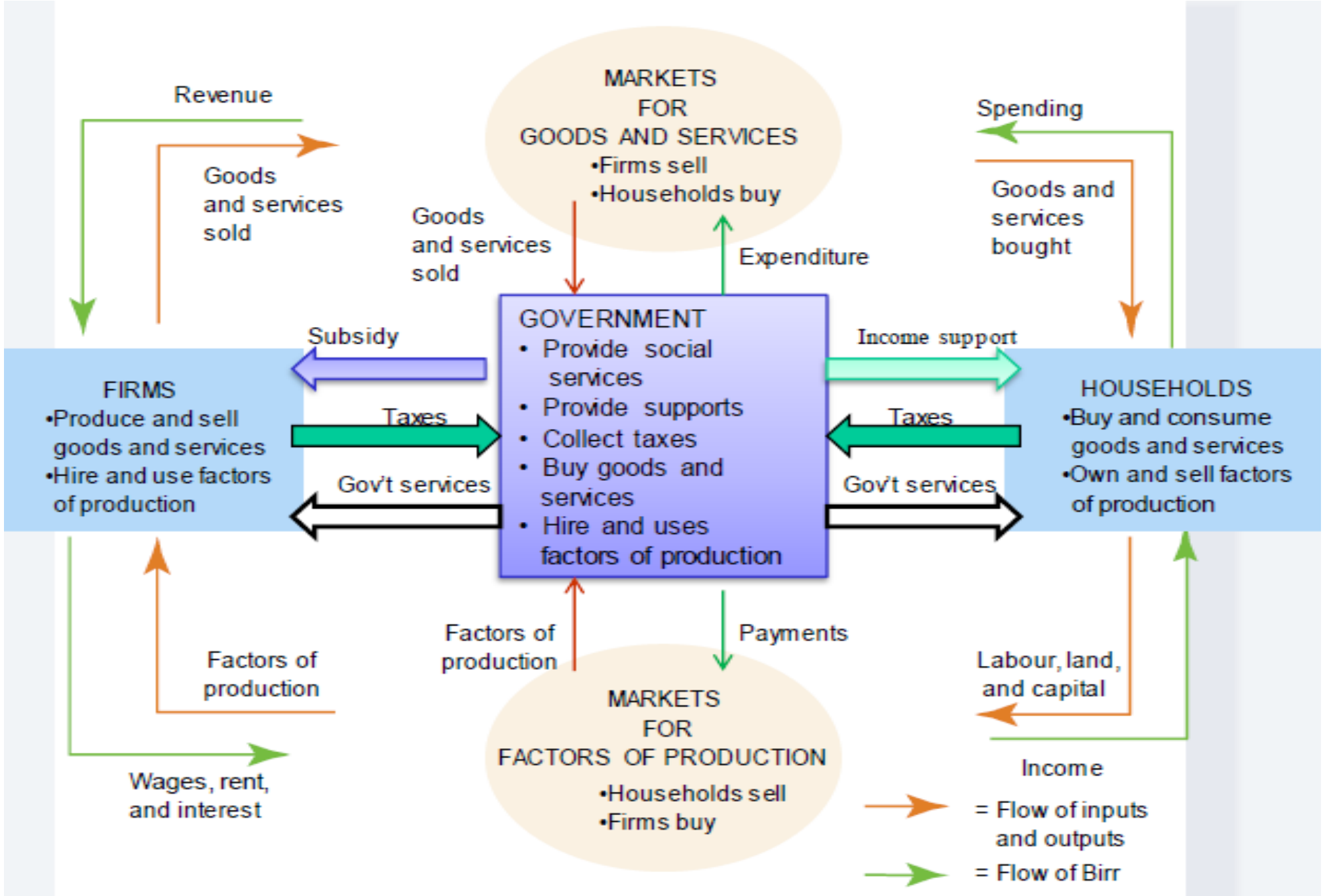
Factor market (input market): it is a market where economic units transact/exchange factors of production (inputs). In this market, owners of resources (households) sell their resources to business firms and governments.

The **circular-flow diagram** is a visual model of the economy that shows how money (Birr), economic resources and goods and services flows through markets among the decision making units.

Two sector modal



Three sector model



THANK YOU!!!!